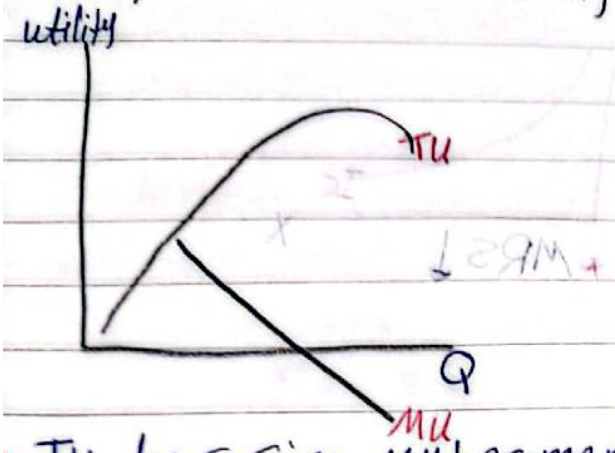
Consumer behavior ⟹ purchase decision

Cardinal ⟹ utility is measured.

⟹ Compare between total utility and marginal utility (in cardinal utility theory).



* TU decreasing, MU become negative
* TU increase at decreasing rate, MU is decreasing.
* TU reach max, MU = 0

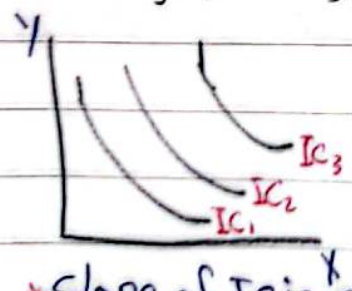
⟹ how to determine equilibrium of consumer according to cardinal

① single good ⟹ $MU_X = P_X$

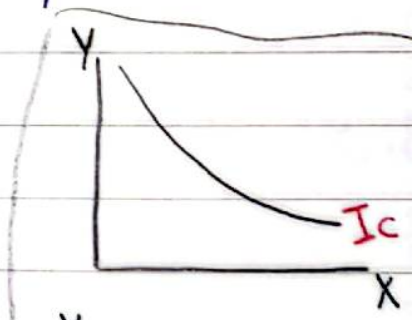
② multible goods ⟹ $\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y} = \text{-----} = \frac{MU_n}{P_n}$

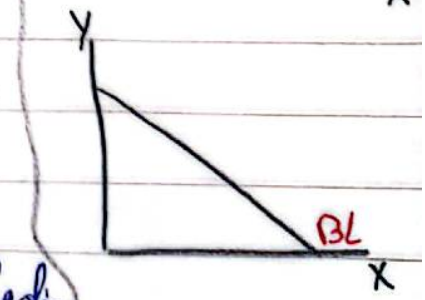
* utility derived from the last pound of each good is equal.

⟹ show graphically indefference map.



# of units of Y that must be given up to ↑ X by 1 unit.





* Slope of IC is called (MRS)
* MRS↓ → IC convex to the origin.
* IC negatively sloped → to keep the same level of satisfaction
* higher IC is better.
* IC never intersect.

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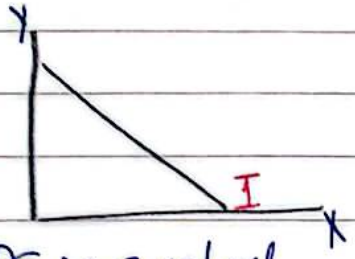
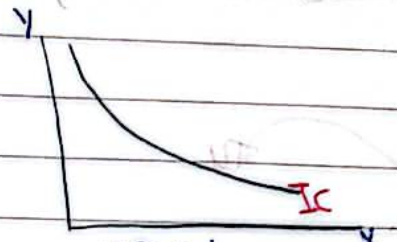
* Show graphically different shapes of indifference curves:

① perfect substitute:



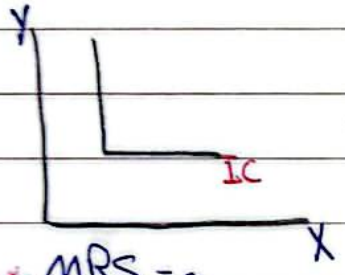
* MRS is constant.

② Close substitute:



* MRS ↓

③ Perfect complement:



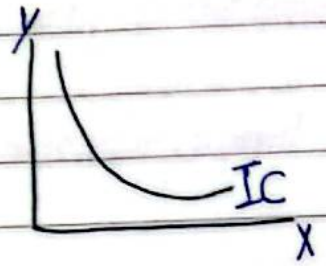
* MRS = 0

* $Qd = f(P_x, P_y, I, taste)$

ordinal theory
(utility is Ranked)

IC

BL (budget line)



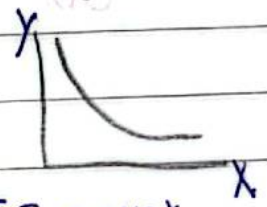
$3Y = 1X$
$2Y = 1X$

- in the same IC, utility is constant.
- higher IC = higher utility.
- IC has negative slope. because utility is constant
- MRS ↓ (marginal Rate of Substitution)
- $MRS = \frac{MUX}{MUY}$ → slope IC

ICs never intersect.

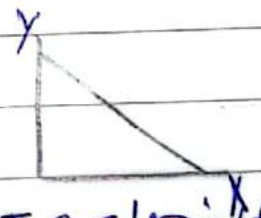
Types of IC
- substitutes
- perfect substitutes.
- Complement

→ substitutes:



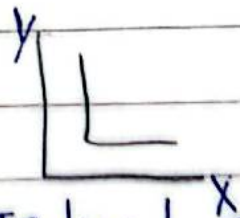
IC convex
MRS ↓

→ perfect substitutes:



IC straight line
MRS constant.

→ Complement:



IC has L shaped
(Right angle)
MRS = 0

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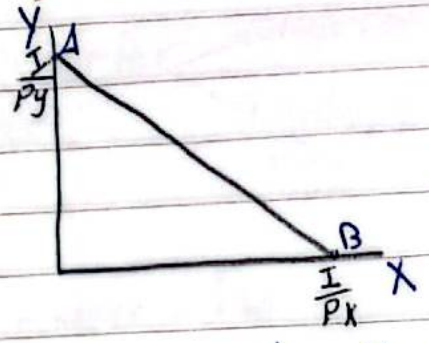
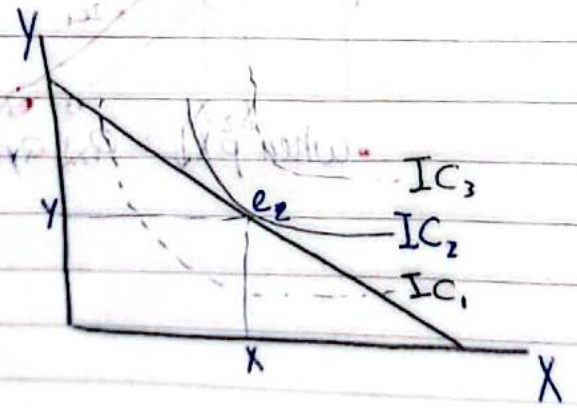
→ budget line (BL)



- A → when consumer buy Y only he will purchase $\frac{I}{P_y}$ units from Y.
- B → when consumer buy X only he will purchase $\frac{I}{P_x}$ units from X.

* $I = P_x Q_x + P_y Q_y$
* in the same BL I is constant.
* slope of BL $= \frac{\Delta Y}{\Delta X} = \frac{P_x}{P_y}$
* Factors affecting BL
  - income
    - ↑ → shift outward.
    - ↓ → shift inward.
  - Price of X
    - ↑ → BL Rotate inward
    - ↓ → BL Rotate outward

* equilibrium



* at equilibrium, consumer by choosing this point, he maximizes his utility give his constraints < income, P_x, P_y

* $MRS = \frac{MU_x}{MU_y} = \frac{P_x}{P_y}$

2 Conditions
- BL is tangent to IC $\frac{MU_x}{MU_y} = \frac{P_x}{P_y}$
- IC must be convex MRS↓

$* PED = \frac{\% \Delta Q_{dx}}{\% \Delta P_x}$

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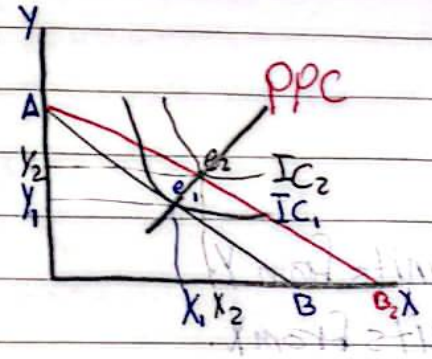
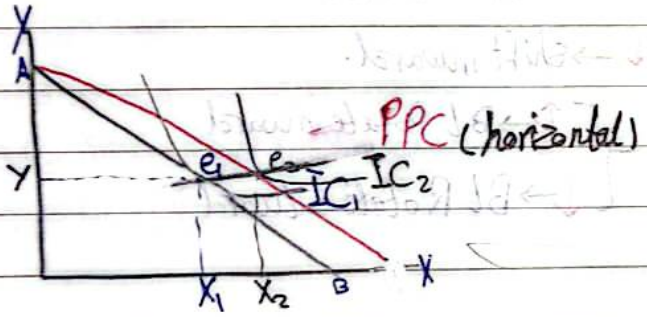
* PPC ⟹ show how consumer's choices changes when price of X change ($P_X \downarrow$)

① incase of inelastic D_x:



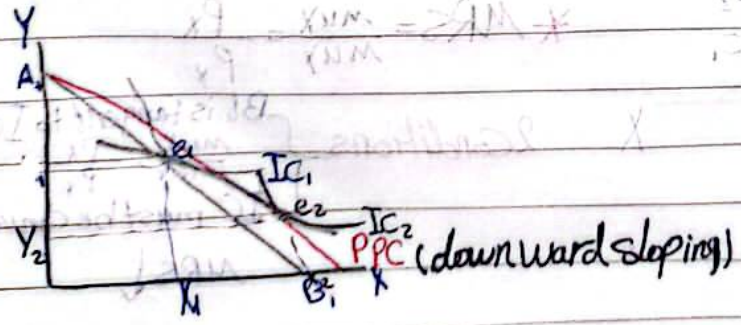
- $\% \Delta P_x > \% \Delta QD_x$
- when $P_X \downarrow$: $Q_X \uparrow$ $Q_Y \uparrow$

② unitary elastic D_x:



- $\% \Delta P = \% \Delta QD_x$
- when $P_X \downarrow$: $Q_X \uparrow$ $\overline{Q_Y}$

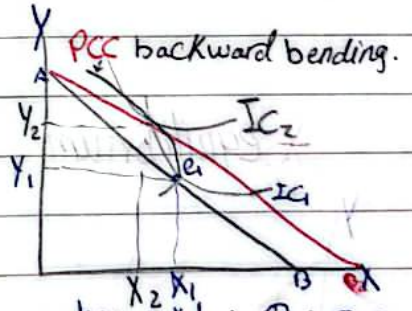
③ incase of elastic D_x:



- $\% \Delta P < \% \Delta Q_D$
- when $P_X \downarrow$: $Q_X \uparrow$ $Q_Y \downarrow$

④ incase of giffen good x

as $P_X \downarrow$ $Q_X \downarrow$



- when $P_X \downarrow$: $Q_X \downarrow$ $Q_Y \uparrow$